NATIONAL DEFENCE UNIVERSITY–KENYA

DEFENCE COLLEGE OF HEALTH SCIENCES

RESEARCH PROPOSAL

SALAMA CHECK-IN: AN ADAPTIVE, CONFIDENTIAL DIGITAL MENTAL HEALTH SERVICE WITH NURSE CALL-BACK AND EARLY WARNING FOR THE KENYA DEFENCE FORCES COMMUNITY AND THE DEFENCE COLLEGE OF HEALTH SCIENCES: CO-DESIGN, THREE-MONTH PILOT AND COSTED ADOPTION PLAN

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ABSTRACT

Mental health problems among soldiers are common, treatable and badly under-reported. Across military studies, about six in ten personnel with a mental health problem do not seek help, mostly from fear of being seen as weak and fear that leadership will treat them differently (Sharp et al., 2015). In Kenya, long deployments, a culture of self-reliance, few specialists and doubt that a clinic visit will stay private add to the problem. Serving personnel and their families now face new and visible pressures: reports of suicides and violence at home, and money problems that include borrowing from informal lenders who hold ATM cards or military identity cards as security. At the Defence College of Health Sciences (DCHS) of the National Defence University–Kenya (NDU-K), staff report that more students are leaving programmes with mental-health-related difficulties, yet the college has no routine, confidential way to find and support students early.

This study will **adapt, pilot and evaluate Salama Check-In** with students who are already enrolled at DCHS. It is an adaptive, confidential digital service. A student completes a private check on a phone and sees a result only they can read (green, amber or red). The student can choose to be contacted by a nurse within a set time, be guided to money, family and welfare help, ask a nurse an anonymous question, or use short micro-lessons. A weekly pulse and group-level early warning let the service change as pressures change. The service is a **support and early-help tool, not an assessment**: it does not affect marks, progression, discipline or sponsorship, and lecturers and commanders never see individual answers. A student may choose to share only a colour with the college student-support office. Because DCHS students will be posted to units and deployed, the same private code and app are designed to go with them after the course.

The design is a pilot implementation study with mixed methods, built to be low-cost and to be completed by one student. Students are invited through a free web link, QR code or printed copy. The service runs at DCHS for at least four weeks, and up to three months, with a nurse on call in stated cover hours and a written safety protocol. Questions cover wellbeing (PHQ-9, GAD-7, PC-PTSD-5, AUDIT-C, resilience), life pressures, money and informal borrowing, stressors in the student's own words, **thoughts of leaving the course and why**, and open boxes for suggestions that will guide future adjustments. Evaluation is measured inside the service: uptake, time to first contact, referral completion, usability, perceived confidentiality and help-seeking intention. The study also describes how wellbeing changes over the pilot and explores whether baseline signals are associated with later early exit, so that support can be targeted and not so that anyone is excluded. A short expert review by a few DCHS staff checks the wording of the questions. Data are held in a database that exports to jamovi.

A guaranteed core and stretch components. The core is the DCHS pilot with enrolled students, with the expert review of questions and testing of the questions with a small group of students. Stretch components, added only when agreed in writing, are applicants, serving units and registry outcome linkage.

The pilot will show uptake and feasibility. It will not show clinical improvement. Expected outputs are a tested service, a costed adoption plan, a policy brief for DCHS and the Medical Services, and evidence on what students need in order to trust and use a digital wellbeing service.

Keywords: military mental health; digital health; student wellbeing; screening; referral; confidentiality; financial stress; Kenya Defence Forces; implementation pilot.

CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Soldiers carry a special kind of work stress. They are away from family for long periods, they are exposed to danger and loss, and they are trained to keep going no matter how they feel. Military populations therefore carry a heavy load of depression, anxiety, post-traumatic stress disorder (PTSD) and harmful alcohol use. A classic study of United States soldiers returning from Iraq and Afghanistan found that mental health problems were common and that many affected soldiers did not seek care because of barriers, with stigma prominent among them (Hoge et al., 2004). A systematic review and meta-analysis of 20 papers reported that about 60% of military personnel with mental health problems do not seek help, and that the two most endorsed stigma concerns were “my unit leadership might treat me differently” (44.2%) and “I would be seen as weak” (42.9%) (Sharp et al., 2015).

African military populations are less studied, but the evidence points the same way. A cross-sectional study of 612 admitted soldiers in Ethiopia found a PTSD prevalence of 21.9%, with higher odds among those with many deployments, heavy combat exposure, poor social support and depression (Tedla et al., 2024). In Uganda, 13% of 247 soldiers retiring from service screened positive for PTSD, and depression and hazardous drinking strongly raised the odds (Bigirwa et al., 2025). A review of West African military studies found only three eligible papers, all from Nigeria and all about substance use, which shows how thin the evidence is (Asare-Doku et al., 2021).

Kenya has been deeply involved in regional security operations, including operations in Somalia since 2011. A Kenyan thesis of 226 soldiers who had served under the African Union mission in Somalia reported that 9% had symptoms consistent with PTSD, 16% showed problematic substance use, 23% reported psychosocial effects of deployment, and only 8% knew that mental health support was available to them (Nyanumba, 2026). The same study linked these gaps to stigma and weak institutional services and recommended screening before and after deployment. A related study of widows of soldiers killed in Somalia found that the coping support offered did not meet their psychosocial needs (Kahiri et al., 2026).

1.1.1 New and emerging pressures on personnel and families

Three pressures have become more visible and are the reason this proposal has been reshaped.

- **Suicide and mental-health-related crises.** Media reports of suicides and violent incidents involving serving personnel have increased public concern. These reports are anecdotal. No official published statistics on suicide in the Kenya Defence Forces (KDF) were found while preparing this proposal, which is itself a finding: the service needs a safe way to count and respond to need without naming people.

- **Violence at home.** Systematic reviews of military populations estimate that about 13% perpetrate and about 21% experience recent intimate partner violence in population-based surveys (Cowlishaw et al., 2022), and that about 26% of military personnel report past-year physical perpetration in a pooled analysis of 28 studies (Kwan et al., 2020). Kenyan figures for military families are not available. Stress, alcohol use, long separations and money worries are common precursors, and families are often the first to see a problem.
- **Money problems and informal lending.** Reports describe personnel borrowing from informal daily, weekly or monthly lenders ("shylocks") at very high cost, handing over bank cards, phones or military identity cards as security, and sometimes reporting a card as lost while a lender holds it. This is a safety, discipline and mental health problem together. No published Kenyan data on it were found, so the study asks about it directly, anonymously and non-judgementally, and asks people to name in their own words what weighs on them most.

1.1.2 The Defence College of Health Sciences and its enrolled students

DCHS trains health professionals, including the Bachelor of Science in Nursing with Military Health, within NDU-K. Students move from school, civilian life or service into a demanding mix of academic work, clinical placements, military discipline and time away from home. The students who leave are the ones who are already enrolled, so this is where early help can make a difference. International reviews show that emotional and psychological distress, heavy workload, weak institutional support and financial strain are repeatedly linked to attrition in undergraduate nursing, which ranges from about 10% to over 30% across countries (Sheikoleslami et al., 2025). In Kenya, a survey of 1,424 university students at one public university reported that about three in ten screened positive for a mental health condition and that family support and age were associated with risk (Mugotitsa et al., 2025). DCHS staff report that more enrolled students are leaving with mental-health-related difficulties. The college does not yet have a routine, confidential way to identify support needs early. DCHS students are also future military health workers: after the course they will be posted to units and deployed, and will care for troops under the same pressures, so a service that supports them now can continue to support them, and those they serve, later.

1.1.3 Why a digital service, and why a service rather than a survey

Digital tools can offer privacy, reach and low cost, and early evidence from Kenya and the region is encouraging (Khakali et al., 2026; Kaigwa et al., 2022; Onu & Onyeka, 2024). But a screening tool that is opened once and forgotten changes nothing. The study therefore treats the platform as a small **service**: a private check, a result that can be used, and a real nurse call-back within a set time, with useful features that bring people back (micro-lessons, an anonymous ask-the-nurse box, a welfare navigator for money, family and housing problems, buddy reminders and peer champions). The evaluation sits inside the service and measures

what people do, such as bookings made, time to first contact and referrals that reached a visit, as well as what they say.

1.2 Problem Statement

Students at DCHS face rising and changing pressures, including workload, money problems that include informal lending, family problems, alcohol use and distress, yet most do not seek help, mainly from fear of stigma and loss of confidentiality. Some leave the programme. The college has no early, confidential, voluntary way to see who is struggling and what is weighing on students, no routine way to see which pressures are growing, and no service that adapts. Using a wellbeing score to judge or select students would be unreliable, because people who fear consequences tend to under-report, and unfair, because it can discriminate against people with mental health conditions. What is missing is a confidential, adaptive, evaluated service that connects students to timely human help, protects them from harm, and gives DCHS and the Medical Services evidence on whether it works and what it costs.

1.3 Justification

The study matches a pressing need and an adoption route. First, it offers DCHS and the Medical Services something concrete to adopt, a working and costed service rather than a report to shelve. Second, it responds to the reviewer's guidance that the project must be a small service evaluated as it runs. Third, it is cheap: it uses free web tools, open-source software, existing nurses and low data use, so that scale-up does not depend on new staffing. Fourth, it protects people: individual scores stay on the phone, sharing with the college is a separate, withdrawable choice limited to a colour, and the service is explicitly not an assessment or discipline tool. Fifth, it is designed to change with the situation, so that new pressures, such as a spike in money stress after a fees deadline or distress after a loss, are seen early and answered. Sixth, it listens: open suggestion boxes record what students want changed, so that the service can be adjusted after the pilot.

1.4 Aim, Objectives, Research Questions and Hypotheses

1.4.1 Broad objective

To adapt, pilot and evaluate an adaptive, confidential digital wellbeing check with nurse call-back and welfare navigation for enrolled DCHS students, and to produce a costed plan for its adoption.

1.4.2 Specific objectives

1. To adapt the service and its questions to DCHS students, using a short expert review of the wording, testing with a small group of students, and open suggestion boxes that guide future adjustments.

2. To run a pilot of the service at DCHS, of at least four weeks and up to three months, with a nurse on call in stated cover hours and a written safety protocol.
3. To measure uptake, time to first contact, referral completion (visits reached), usability, perceived confidentiality and help-seeking intention.
4. To describe the stressors that students name themselves, across work and the course, family and home, and money (including informal borrowing and the holding of identity cards as security), together with contact with civilian police, and the reasons students give for thinking about leaving the course.
5. To describe how students' wellbeing changes over the pilot, and to explore whether baseline wellbeing, stressors and thoughts of leaving are associated with later early exit (deferral, withdrawal or discontinuation), in order to target support and not to exclude anyone.
6. To produce a costed adoption plan and a policy brief for DCHS and the Medical Services.

Table 1.1 Objectives made specific and measurable. The targets are pilot targets, to be confirmed with the supervisor and the steering group before launch.

Objective	Measure	Target	By
1. Adapt	Wording reviewed by five DCHS staff; tested with 10 to 15 students	All items reviewed; at least 90% of testing students find the questions clear; unclear items rewritten	Month 3
2. Pilot	Pilot run with a named nurse, a back-up and a signed safety protocol	At least four weeks live; protocol signed before the first student joins	Months 4 to 6
3. Measure	Complete check-ins; SUS score; time to first contact; visits reached	At least 100 complete check-ins (below that, counts only); SUS above 68; 80% of same-day requests contacted within 8 hours and 80% of routine requests within 7 days	Month 6
4. Describe pressures	Share naming each pressure, by group of 10 or more; share choosing Prefer not to say	Every pressure reported with its Prefer-not-to-say share; no cell under 10 shown	Month 7
5. Change and early exit	Repeat check-ins per student; later deferral or withdrawal	Change described for students with two or more check-ins; association explored, no prediction claim	Month 7
6. Adoption	Costed plan and policy brief delivered	Both delivered to DCHS and the Medical Services	Month 8

1.4.3 Research questions

1. What changes do students and expert reviewers suggest to the questions and the service, and what would make students trust it?
2. Is the service used, and do students who ask for help reach a nurse and a visit within the targets set?

3. How usable is the service, and how confidential do students believe it is?
4. What pressures do students name, how common are informal borrowing and held ID cards, and what reasons do students give for thinking about leaving?
5. How does wellbeing change over the pilot, and are baseline signals associated with later early exit?
6. What would it cost DCHS and the Medical Services to adopt the service?

1.4.4 Hypotheses (feasibility pilot, to be tested where numbers allow)

- **H1.** Students who report often thinking about leaving the course will have higher mean PHQ-9 and money strain scores than students who never think about it.
- **H2.** The mean System Usability Scale score will be above 68.
- **H3.** At least 80% of same-day requests will be contacted within 8 hours of receipt or of cover starting, whichever is later (see cover hours in Chapter Three), and at least 80% of routine requests within 7 days (targets to be confirmed in the pilot).
- **H4.** Perceived confidentiality and help-seeking intention will be higher at the end of the pilot than at baseline.
- **Exploratory.** Money strain, informal borrowing, higher symptom scores and frequent thoughts of leaving at baseline are associated with early exit. No hypothesis about prediction accuracy is set because numbers are small.

1.5 Significance of the Study

For students, a private first step and a real answer. For nurses and the college, a way to see where need is growing and to respond in time, with a costed route to adoption. For DCHS, an evidence-based, voluntary student-support tool that can be extended to other NDU-K colleges and, after graduation, to the units its graduates serve. For the field, one of few evaluations of a digital wellbeing service in an African military health college, and the first to measure money and loan stress alongside students' reasons for thinking about leaving.

1.6 Scope and Delimitations

The study covers enrolled DCHS students in all programmes and years who are aged 18 or older and agree to take part, over a pilot of at least four weeks and up to three months, with outcome follow-up to the end of the term if agreed. It does not diagnose, treat or measure clinical recovery. Findings will not be generalised to all of the KDF or to other colleges. Applicants, serving units, family members and a comparison unit are possible later extensions and are outside the guaranteed core. Students under 18 are excluded.

1.7 Operational Definitions

Term	Meaning in this study
Wellbeing check	A private set of questions on mood, worry, trauma reactions, alcohol, money, life pressures and coping, scored on the phone.
Colour band	Green (no support need flagged), amber (some strain or struggling) and red (needs care now), derived from the four-step wellness zones. A screen, not a diagnosis.
Pilot service	The period (at least four weeks, up to three months) in which the check, nurse call-back, navigator, lessons and ask-the-nurse box are offered and evaluated.
Time to first contact	Hours from a request being sent to a nurse’s first status change or reply, recorded by the database.
Referral completion	A request that reached a completed visit.
Informal lender	A lender outside a bank, SACCO or welfare scheme, for example a daily, weekly or monthly lender.
Early exit (attrition)	Deferring, withdrawing or being discontinued from the programme.
Thinking of leaving	A student’s own answer to whether, in the last three months, they have thought about leaving or deferring the programme.
Honest broker	A designated college registry officer, who is not a lecturer, examiner or member of any discipline panel, and who supplies only a private code and a later outcome to the research team.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews evidence on military mental health and help-seeking, violence at home and suicide, financial stress, digital mental health in low-resource and military settings, student mental health and attrition, and the problems of using self-report screening where people fear consequences. It then sets out the legal and ethical context, the theoretical and conceptual frameworks, and the gaps the study addresses. Sources were identified in Google Scholar, PubMed and publisher sites, and from Kenyan university repositories, using terms for military, mental health, stigma, digital health, Kenya, student attrition and screening. Media reports are used only to describe public concern and are labelled as anecdotal.

2.2 Mental health and help-seeking in military populations

Depression, anxiety, PTSD and harmful drinking are common in service members, and many do not seek help (Hoge et al., 2004). In a meta-analysis, stigma about being seen as weak and fear of different treatment by leadership were the most endorsed barriers, with about 60% of affected personnel not seeking help (Sharp et al., 2015). Stigma has a small to moderate effect on help-seeking in general populations (Clement et al., 2015). African military data are limited and show high PTSD and substance use in Ethiopia, Uganda and Kenya (Tedla et al., 2024; Bigirwa et al., 2025; Nyanumba, 2026), with little knowledge of available support among Kenyan soldiers who served in Somalia. Okulate and Jones (2006) linked PTSD, guilt and substance use in Nigerian veterans. Families carry part of the burden: Kahiri et al. (2026) found that support to widows of Somalia casualties fell short of need. The Kenya Mental Health Policy 2015–2030 calls for accessible, community-based services and for mental health to be integrated into primary care, which suggests a role for nurse-led first contact (Ministry of Health, 2015).

2.3 Violence at home and suicide among military families

Pooled estimates for recent intimate partner violence in military and veteran populations are about 13% for perpetration and 21% for victimisation in population-based surveys (Cowlishaw et al., 2022), and 26% of military personnel report past-year physical perpetration in a meta-analysis of 28 pooled studies (Kwan et al., 2020). Both reviews are mostly from high-income countries, and measurement differs between studies, so the figures indicate seriousness, not a Kenyan rate. For suicide, no official KDF statistics were found. Public discussion rests on media reports, which should be read as signals of concern and not as rates. A service that asks about thoughts of self-harm, safety at home and worry about one’s own anger or drinking, and that responds on the same day to high risk, is consistent with this evidence, provided it protects people from being identified.

2.4 Financial stress and informal lending

Financial strain is associated with depression, anxiety, relationship conflict and, in some studies, suicidal thinking in general populations, and money worries are a recognised reason for leaving educational programmes (Sheikoleslami et al., 2025). Kenya's credit market includes many digital and informal lenders. Regulation of non-deposit-taking credit providers by the Central Bank of Kenya was introduced by recent law and regulation, and the details should be confirmed at the time of the study. In the military setting, informal lenders who keep an ATM card, phone or military identity card as security create specific risks: loss of control over salary, harassment, pressure to report a card as lost, and discipline problems. No published Kenyan study on this was found. This is a gap, and a reason to measure it carefully and ethically.

2.5 Digital mental health in military and low-resource settings

Reviews of digital resilience and wellbeing interventions for military members show promise but weak evidence, variable quality and low sustained engagement (Allen et al., 2025; Tam-Seto et al., 2018). In Kenya, a mixed-methods pilot found a mobile app and wearable feasible and acceptable for collecting mental health data from health care workers (Khakali et al., 2026), and a Kenyan telepsychiatry study found it acceptable (Kaigwa et al., 2022). A scoping review of digital psychiatry in Nigeria and a review of digital technologies for maternal and child health in low- and middle-income countries note that connectivity, trust, data protection and integration with existing services determine success (Onu & Onyeka, 2024; Till et al., 2023). The World Health Organization advises that digital health supplements and does not replace health system functions and calls for attention to privacy and equity (World Health Organization, 2019). Standards-based design, such as SMART on FHIR, supports later integration with health records (Mandel et al., 2016). The lesson is that engagement features, human follow-up and trust matter more than the technology.

2.6 Student mental health and attrition

A scoping review of 19 studies on undergraduate nursing attrition found four themes: institutional and social support (the most frequent), academic factors, personal factors including emotional distress and family responsibilities, and economic challenges. Retention strategies included early identification of at-risk students, mentorship, emotional and psychological support, financial assistance and peer networks (Sheikoleslami et al., 2025). In Kenya, about 30.9% of 1,424 university students screened positive for a mental health condition, and strong family support was associated with lower odds (adjusted OR 0.47) (Mugotitsa et al., 2025). Together these support an early, voluntary, support-focused approach at DCHS and point to money and family support as pressures worth measuring.

2.7 Self-report screening where people fear consequences

Self-report measures such as the PHQ-9 and GAD-7 are validated for screening, including in East Africa (Kroenke et al., 2001; Spitzer et al., 2006; Gelaye et al., 2013; Mwangi et al., 2020). They are screens and not diagnoses. When answers can have consequences, people have a reason to answer in the way that protects them. In job-selection research, applicants score higher than non-applicants on desirable personality traits by about 0.45 of a standard deviation for conscientiousness and emotional stability (Birkeland et al., 2006). This is evidence from selection, not from mental health screens, so its size here is an assumption. Enrolled students fear not rejection but consequences for marks, progression, discipline or sponsorship, and in a military college they may fear that commanders will treat them differently, a concern that is the most endorsed in military help-seeking research (Sharp et al., 2015). The best protection is to remove the consequences and to make that credible: the check is voluntary, anonymous, scored on the phone and never seen by staff who teach, mark or discipline students. A wellbeing score used to judge students would be biased towards under-reporting, would discourage honesty and help-seeking, and would risk discrimination against people with mental health conditions. A voluntary, support-focused use is safer, more honest and consistent with the aim of keeping students in the programme. The study checks this by asking students how much they feared consequences while answering and whether they answered honestly.

2.8 Legal and ethical context

- **Constitution of Kenya, 2010.** Article 27 guarantees equality and freedom from discrimination, including on grounds of disability, and Article 31 protects privacy.
- **Mental Health Act (as amended in 2022).** Persons with mental illness are protected from abuse and any form of discrimination, and information about care and treatment is confidential, with disclosure allowed only in listed cases such as where required by law or necessary for treatment or the person’s best interest (Mental Health (Amendment) Act, 2022).
- **Persons with Disabilities Act.** Prohibits discrimination in education and employment on the ground of disability (Republic of Kenya, 2003, as amended).
- **Data Protection Act, 2019 and General Regulations, 2021.** Health data is sensitive personal data, with duties of lawful basis, minimisation, security, data protection impact assessment and registration with the Office of the Data Protection Commissioner. Regulation 26 limits processing for certain health purposes to servers or a serving copy in Kenya (Republic of Kenya, 2019, 2021).
- **Research ethics.** The Nuremberg Code, Declaration of Helsinki and Belmont Report set out respect for persons, beneficence, non-maleficence and justice (World Medical Association, 2013; National Commission for the Protection of Human Subjects, 1979). Kenyan research requires institutional ethics review and a NACOSTI licence.

2.9 Theoretical framework

Three frameworks guide the work. **Design science** (Hevner et al., 2004) treats the platform as an artifact that is built, tested and improved against a real need. **UTAUT** (Venkatesh et al., 2003) explains acceptance through performance expectancy, effort expectancy, social influence and facilitating conditions, and is used to build the acceptance items. **Implementation outcomes** (Proctor et al., 2011) and **RE-AIM** (Glasgow et al., 1999) define what the pilot measures: reach, adoption, acceptability, feasibility, fidelity and, later, cost and sustainability. The four-zone wellness scale is adapted from the Mental Health Continuum Model used by the Canadian Armed Forces (National Defence, 2025).

2.10 Conceptual framework

Pressures and cues (academic load, money and loans, family and home, being away, news and loss, sleep and alcohol) feed into a private wellbeing check. The check gives a result and a colour band that only the person sees, and the person chooses what to do next: lessons, an anonymous question, a nurse call-back, the welfare navigator or a referral. Outcomes are uptake, time to contact, referrals that reached a visit, usability, perceived confidentiality and help-seeking intention. Stigma, trust in confidentiality and perceived stakes moderate the path. An adaptive loop senses change, detects group-level signals, applies pre-agreed rules and records every change (Figures 1 and 3).

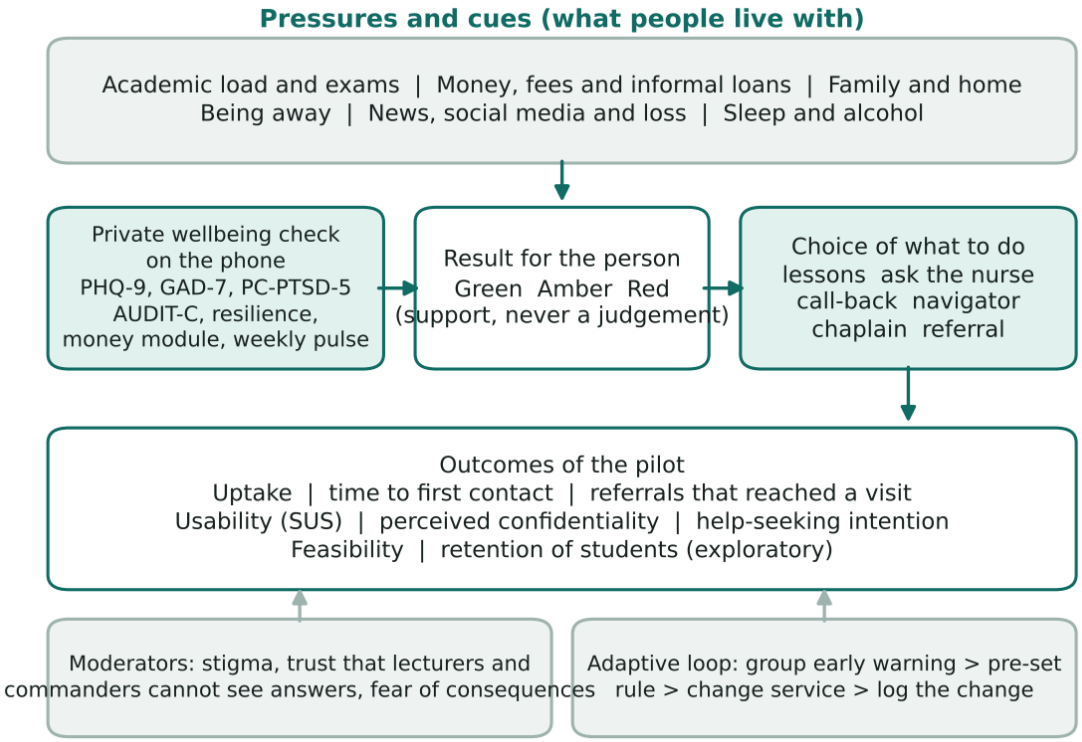


Figure 1. Conceptual framework

2.11 Gaps addressed by this study

- No tested digital mental health service for KDF personnel and families in Kenya, evaluated by what people do as well as what they say.
- No published data on informal lending, the holding of identity cards as security and false lost-card reports among service members, and no anonymous, non-punitive way to ask about them.
- No evidence on how fear of consequences colours the answers of students in a military health college, and on what students themselves give as reasons for thinking about leaving.
- Few services that adapt to changing pressures and show their changes openly.

CHAPTER THREE: RESEARCH METHODOLOGY

3.1 Study Design

The study is a **pilot implementation study with mixed methods**. A small service is built from open-source tools, adapted with students and a short expert review, run at DCHS, and evaluated as it runs. The evaluation uses the RE-AIM dimensions of reach, adoption, implementation and maintenance (Glasgow et al., 1999) and the implementation outcomes of acceptability, adoption, appropriateness, cost, feasibility, fidelity and penetration (Proctor et al., 2011). Design-science thinking (Hevner et al., 2004) guides the build-test-improve cycle. Effectiveness in the clinical sense (fewer cases of illness) is **not** claimed: a pilot of a few weeks in one college can show whether the service is used, trusted and acts on time. It cannot show that mental health improved. This limit is stated here and again in 3.13.

Scope and priorities. The study is built in layers so that one student can finish it at low cost. The **guaranteed core** is the pilot with enrolled DCHS students, an expert review of the questions, and testing of the questions with a small group of students. **Stretch components** are applicants, serving units, family members, a comparison group, and linkage to registry outcomes. Stretch components start only when their written agreements are in place and never delay the core (3.3.2). No Delphi panel, key informant interviews or group discussions are planned, because they are costly and not needed for the pilot. Instead, **open suggestion boxes** in the questionnaire and a short expert review give room for opinion and for future adjustment.

3.2 Study Area

The study is carried out at the **Defence College of Health Sciences (DCHS)**, a college of the National Defence University–Kenya (NDU-K), at latitude **1.3124° S and longitude 36.8043° E** in Nairobi (Figure 5). The site is in Nairobi County, a little over 3 km south-south-west of the city centre. DCHS trains health professionals, including nursing students in the Bachelor of Science in Nursing with Military Health, who come from across Kenya and live and study in a setting that combines academic work, clinical placements and military routine. Students are reached on their own phones, so the study does not depend on the college network.

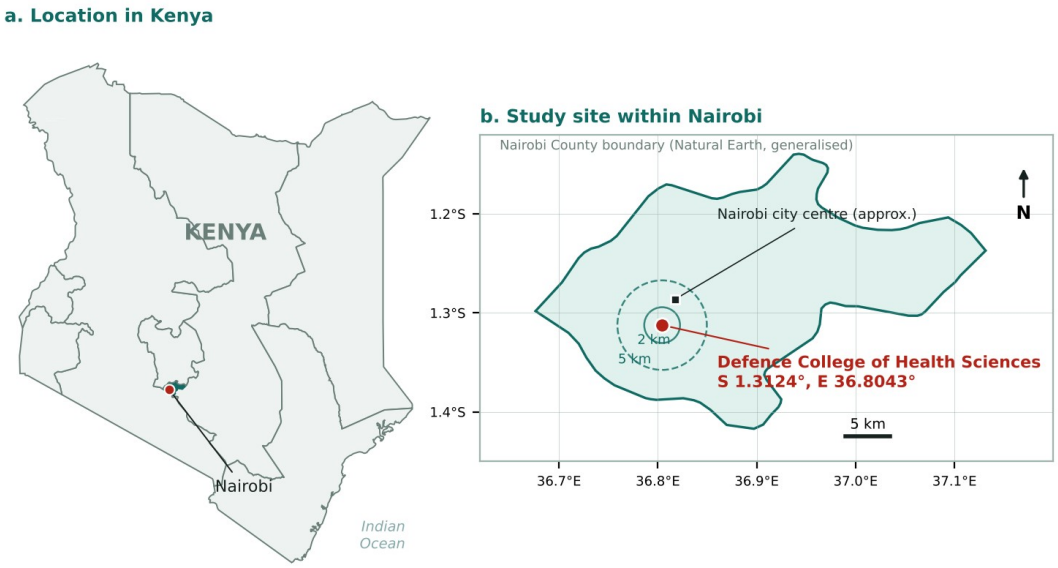


Figure 5. Study area: DCHS in Nairobi (S 1.3124°, E 36.8043°). Boundaries: Natural Earth (generalised). Distance rings at 2 km and 5 km.

3.3 Study Phases, Participants and Recruitment

3.3.1 Phases and timeline

The work takes eight months from approval of this proposal (Figure 4 in Chapter Five). Each phase has an end point that must be met before the next begins.

Phase	Months	Activities	Objective	Output
0. Approvals and agreement	1–2	University ethics review; KDF-ISERC approval (no fee) for research within the military establishment; NACOSTI licence; letter of support from DCHS (results not used for marks, progression or discipline; call-back and referral route; where data are kept); data protection impact assessment	All	Approvals; letter of support
1. Adapt and test	2–3	Expert review of questions (five DCHS staff); testing with 10 to 15 students (3.9.1); build and technical checks (3.9.2); safety protocol; nurse rota; peer champions briefed	1	Revised tools; protocol; go or no-go
2. Pilot	4–6	Live at DCHS for at least four weeks and up to three months: baseline check, weekly pulse, follow-up check about every four weeks, nurse on call in cover hours, group early warning, engagement features, open suggestions	2, 3, 4, 5	Service log; check-ins; call-back records
3. Follow-up and synthesis	6–8	End-of-pilot feedback; registry outcome file if agreed; analysis in jamovi; costed adoption plan; policy brief; dissertation	3, 5, 6	Findings; adoption plan; brief

3.3.2 Guaranteed core and stretch components

Component	Status	Depends on	If it cannot proceed
Expert review of questions and testing with students (Objective 1)	Core	Ethics approval	None; done on paper or on phones
Pilot with enrolled DCHS students, nurse on call (Objectives 2, 3, 4)	Core, at least four weeks	DCHS letter of support and a nurse in cover hours	Run the survey-only version with engagement features (3.12)
Change over the pilot, and early exit (Objective 5)	Core for change; early exit exploratory	Registry outcome file for early exit	Report change over the pilot, and self-reported thoughts of leaving, without registry outcomes
Costed adoption plan and policy brief (Objective 6)	Core	Cost log from the pilot	Use planning assumptions, labelled as such
Applicants (before enrolment)	Stretch	DCHS agreement and ethics amendment	Not done; proposed for the next phase
Serving units, family members, comparison group	Stretch	Written agreement from Medical Services and commanders	Not done; proposed for the next phase

3.3.3 Participants and eligibility

Population: all students enrolled at DCHS in any programme and year. **Inclusion:** aged 18 or older; currently enrolled; has a phone or can use a printed copy; agrees to take part. **Exclusion:** under 18; unable to give consent. A student who is in acute crisis at the time is **not excluded**: the check shows help numbers at once and offers a nurse (3.4.4).

3.3.4 Recruitment and how students are invited

The study is introduced to all enrolled students at a short briefing, delivered by the researcher or a class representative. **Lecturers, examiners and commanders do not recruit and do not see who takes part.** Students receive the link and QR code on a poster and by their own class messaging group. A printed copy (Appendix A) and a sealed box are provided for students who prefer paper. Taking part is **voluntary and anonymous**, and the protections are shown to every student on the first screen of the app and on the information sheet **before** they agree: voluntary; not an assessment; no name or admission number; staff cannot see answers; only the student sees the result; help is close; approved and supervised; and the single limit of confidentiality (immediate danger).

3.4 The Service and How Data Are Collected

The service is **Salama Check-In**, a progressive web app that installs on a phone like an ordinary app, works with poor coverage, and stores answers on the phone until the network returns. It has a private staff console for the nurse, the welfare office and the researcher. Behind it is a database with access rules set at row level, so each staff role sees only what that role needs. Individual scores are calculated **on the phone**; the service receives the answers only after the person sends them.

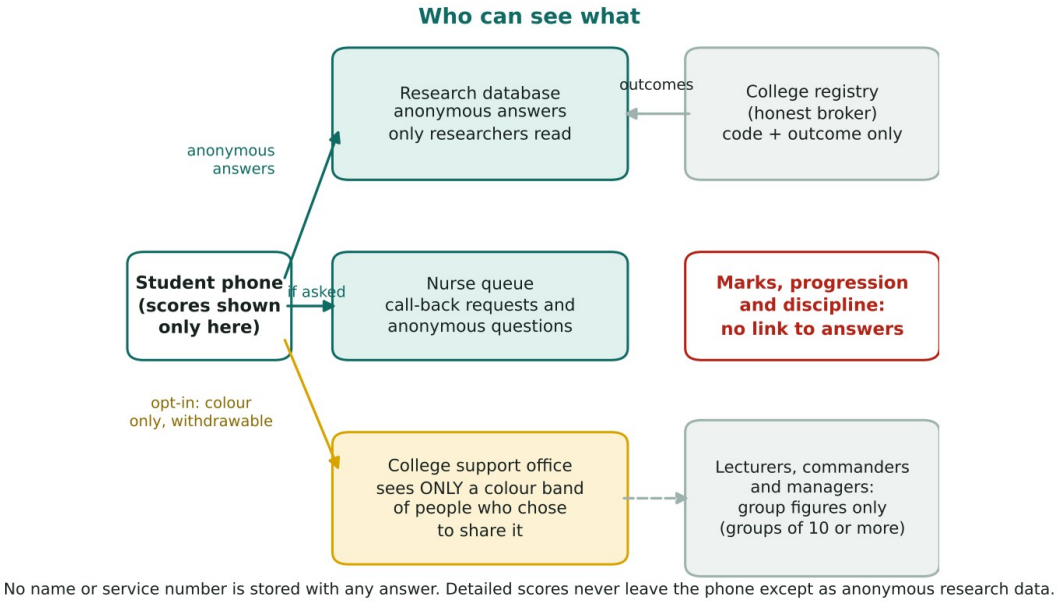


Figure 2. What each person can see: access to individual and group information

3.4.1 What the student sees

- **A private check** of about twelve minutes, with a result shown only on the student's phone as green (coping well), amber (feeling the strain or struggling) or red (needs care now). The questions a student sees depend on earlier answers, so most students see fewer questions than the full list.
- **Help in one tap:** a nurse call-back with a promised time, an anonymous question to the nurse, a welfare navigator for money, housing and family problems, and a **Get help** page with contacts and national lines.
- **Micro-lessons** of two or three minutes (sleep, stress, anger, alcohol, money and loans, family, loss, exams, first weeks at college).
- **Engagement features:** a weekly pulse, buddy reminders, and peer champions who show others how the service works and what stays private.
- **Safety features:** a Quick Exit button on every screen, no name, and neutral reminder messages.
- **A guide** that points to lessons and people. The on-phone guide matches a question to a lesson and shows the safety page first when it detects crisis or danger words in

English or Kiswahili. An AI guide is built but **switched off**, and will be used only if ethics and data-protection review approve it. Crisis text is never sent to it.

- **Open suggestion boxes** at the end of the check: what would make the service more useful, what the college or study should change, and whether any question was hard to understand. These responses are read in full and guide future adjustments.

3.4.2 Adaptive features

The service is built to **change as pressures change**, under rules written down in advance (Figure 3):

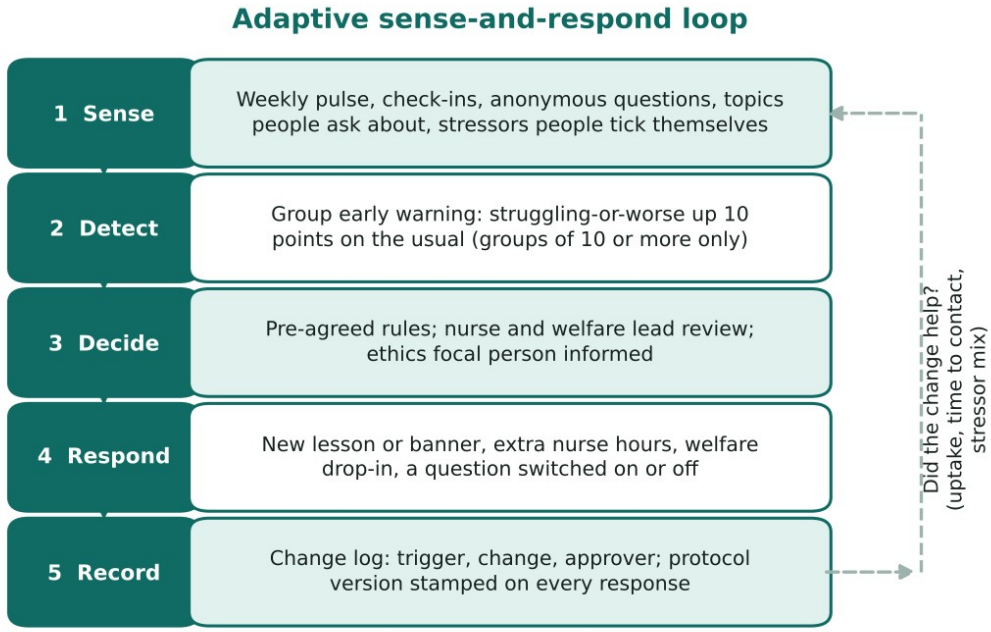


Figure 3. The adaptive sense-and-respond loop

- **Weekly pulse** of eight items, including a cue asking what is weighing on the student this week.
- **Adaptive check:** PHQ-9 and GAD-7 follow a short form. The remaining items are asked only if the first two are above a set level; the suicide item (PHQ-9 item 9) is always asked.
- **Group early warning:** for any group of ten or more, the service compares the current average to the average of the previous four periods. A rise of 10 points or more raises a group alert to the welfare lead and the medical officer, never with names. Groups under ten are hidden.
- **Rules and change log:** each change to a rule, question or message is recorded with a date and reason, and every record carries the protocol version.

3.4.3 Anonymous identity and continuity after college

Each student receives the next anonymous number in the group the first time they join, issued by the database in the order people join (for example STUD-0001, STUD-0002). The number is linked to no name, phone or admission number and is held on the student's phone; if there is no signal at that moment, the phone makes a random code instead. The researcher sees every code in the data but not who holds it. The same code links one student's answers across the baseline check, weekly pulses and follow-up checks, which is what the change analyses need. The code goes with the person after college: in Settings a student records a change of situation (at college, in barracks, deployed, back from deployment, on leave, at home) or role (student to serving member) and keeps their history. The app opens and works without signal after the first load, holds requests until a connection returns, and can offer a prepared text message to the nurse phone where only a mobile network is available. Its fit for deployment conditions is an assumption to be tested, not a result.

3.4.4 Call-back cover and after-hours route

A single nurse cannot cover every hour, so the service states its **cover hours** and never promises more. Proposed cover is **07:00 to 21:00 every day**, shared by two trained nurses with the nurse lead as back-up (to be confirmed with DCHS and the hospital). The times in the protocol are counted from receipt of a request during cover hours, or from the start of the next cover period for a request received outside them. Outside cover hours, the phone tells the student so, shows the duty medical officer, the emergency number and the national lines, and sends any red-band request to the duty medical officer by a route to be agreed. **Overnight nurse cover is not promised.** If no after-hours route can be agreed, the pilot runs on the cover hours alone and the information sheet says so. The nurse lead checks requests per week against nurse minutes at each steering group meeting, and enrolment is paused if the rota cannot meet the targets. Telephone numbers shown in the app during the pilot design are **placeholders (+2547xxxxxxxx)** and are replaced when the service is adopted.

3.4.5 Where the survey is hosted: no college server is needed

The study does **not** depend on DCHS hosting the survey. The app is a set of static files, and the answers go to a separate database. The simplest, free route is:

1. **App link:** the files are published free on **GitHub Pages** (or Netlify or Cloudflare Pages), which gives a permanent web address and a QR code that any student can open on any phone.
2. **Data:** answers are stored in a free **Supabase** project (PostgreSQL), which the researcher reads in the staff console and exports to jamovi as CSV or SPSS files.
3. **Fallbacks:** (a) **KoboToolbox**, a free survey service that hosts the form and gives a link in minutes, with the same questions, used if hosting is delayed; (b) **printed copies** in a sealed box, typed into a prepared data-entry sheet by the researcher.

Study portal and languages. A single portal page gathers the student app (shown in a phone frame with a live record of what is stored and sent), the nurse console, a leadership dashboard, a monthly report, design notes with the printable survey tool, the research survey, the expert-input form, and the data and export page. The portal and the app run in English and Kiswahili with one switch. The Kiswahili wording is a draft: it is checked by a translator, translated back into English by a second person, and, where a validated Kiswahili version of a scale exists (for example the PHQ-9), that wording is used. The dashboard and report show groups of 10 or more only, and totals count visible groups only so that a hidden group cannot be worked out.

What the survey covers. To picture a soldier's life and not only symptoms, the survey covers work and the course, family and home, money (running short, repayments, informal lenders, cards or ID held as security, threats), betting and gambling (how often, needing to bet more, lying about it and borrowing to bet, using the published Lie/Bet screen, wording to be checked against the source), health (PHQ-9, GAD-7, PC-PTSD-5, AUDIT-C, sleep, resilience), safety, belonging, and contact with civilian police (altercations, setting, alcohol, what followed, whether help was available, and fear that a small incident becomes a discipline case). Sensitive items are optional, offer Prefer not to say, and can be switched off in one setting if the ethics bodies advise. No names, places, dates, lender names or service numbers are asked.

DCHS is asked for **support, not hosting**: a letter of support, permission to brief students and put up the poster, nurse time, and later a decision on whether to host a Kenyan copy of the live call-back service. Free hosting services are outside Kenya, so they are used for development, practice data and anonymous survey answers, with the ethics committee's approval. If the live call-back service needs contact numbers, it must run in Kenya, or keep a serving copy in Kenya, to meet regulation 26 of the Data Protection (General) Regulations, 2021. At that point DCHS or the Medical Services would be asked to host it. The budget therefore shows hosting as free for the pilot and notes DCHS hosting as an option for adoption (Chapter Five).

3.5 Sample Size

Because the whole student body is invited (a census), the aim is to obtain as many responses as possible, and the sample size below states the precision needed and what is done if it is not reached. The population N (the number of enrolled students) is taken from the registry and is not yet known. Cochran's formula for a proportion, $n_0 = Z^2pq/d^2$, is used with Z = 1.96 and d = 0.05, and the finite-population correction $n = n_0 / (1 + (n_0 - 1)/N)$ (Cochran, 1977). With p = 0.50, the most cautious assumption, $n_0 = 385$. With p = 0.20, a planning value for the share of students who go on to a visit, $n_0 = 246$.

Table 3.1. Required complete check-ins for a 5-point margin of error, by number of enrolled students (N)

Enrolled students (N)	n at p = 0.50	n at p = 0.20
150	109	94
300	169	136
500	218	166
1000	279	198

Levels. The **ideal** is the figure in Table 3.1 for the true N. The **minimum acceptable** is 100 complete check-ins: with p = 0.20 the margin of error is ±7.8 points. **Below 100** (for example 60, margin ±10.1 points), results are reported as counts with exact confidence intervals only, with no tests, and framed as pilot signals. The pilot is extended week by week, up to three months, until the minimum is met.

Change over the pilot. A paired t-test with a moderate effect (Cohen's d = 0.4), 80% power and 5% two-sided significance needs about **52 pairs**. With 30% loss between baseline and follow-up, about 75 students are needed at baseline. If fewer pairs are available, the effect size and its confidence interval are reported as an estimate and no significance claim is made.

Early exit. A logistic model for early exit needs about ten events per predictor. If 10% to 15% of N leave in the follow-up period, a college of 300 would have about 30 to 45 events, enough for three or four predictors at most. This analysis is therefore **exploratory**, uses a penalised (Firth) correction, and is reported as a signal. It is not used to predict or to select any student.

3.6 Data Collection Tools

All tools are in the **Data Collection Tools Pack** (Appendix A), in the same words the app shows. Questions are written in plain English at about a Grade 6 to 7 reading level, one idea per question, with no double-barrelled items. Standard scales keep their published wording.

Tool	Content	Used for objective
Student wellbeing check (76 questions listed, 108 recorded items; adaptive: 54 always asked)	PHQ-9; GAD-7; PC-PTSD-5; AUDIT-C; life pressures; money, loans and bills; stressors in own words; thoughts of leaving and reasons; resilience (Brief Resilience Scale; Smith et al., 2008); help-seeking intention; home situation; three open suggestion boxes; perceived consequences and honesty	2, 3, 4, 5
Weekly pulse (8 items)	Sleep, stress, money, home, what is weighing on the student this week	2, 4, 5
End-of-pilot feedback (45 items)	System Usability Scale; acceptability (UTAUT); perceived confidentiality; help-seeking intention; open comments	1, 3

Tool	Content	Used for objective
Expert review sheet	Five DCHS staff rate each question for clarity and relevance (1 to 4) and add comments	1
Service log and cost log	Requests, time stamps by the database, visits, nurse minutes, airtime and hosting	2, 3, 6

Scoring uses published cut-offs: PHQ-9 of 5, 10, 15 and 20; GAD-7 of 5, 10 and 15; PC-PTSD-5 of 3 or more; AUDIT-C of 4 or more; Brief Resilience Scale (six items, 6 to 30); System Usability Scale (above 68 is above average). The PHQ-9 and GAD-7 have been validated in East Africa (Gelaye et al., 2013; Mwangi et al., 2020). Zones: **green** (zone 1), **amber** (zones 2 and 3), **red** (zone 4). Questions about money and loans never change the colour; they trigger welfare help.

3.7 Data Management and jamovi Analysis

Answers are stored in a database. Each answer set is a row with a private study code, the form, the date and the protocol version. The researcher exports **flat numeric tables** (one row per answer set) as CSV or SPSS (.sav) files with variable labels and value labels from the codebook, so they open in **jamovi** with the right measurement levels. Ready views give first-versus-later checks for the same code, the cohort with groups under ten hidden, monthly stressor counts and service-time measures. Practice files with invented data are supplied. Backups are exported weekly. For the Kobo fallback, a script adds the scale totals to the Kobo download; for paper copies, a prepared data-entry sheet has one column for each item.

3.7.1 Planned analyses by objective

Objective	Analysis in jamovi
1. Adapt the service	Item-level content validity index from expert ratings (relevant = 3 or 4; I-CVI of 0.80 or more kept); cognitive-interview findings; thematic analysis of the open suggestion boxes (Braun & Clarke, 2006)
2. Pilot	Counts and proportions of requests, completions and visits; run charts by week; fidelity to protocol (share of red and same-day requests contacted within target)
3. Uptake and outcomes	Median and interquartile range of hours to first contact; proportion of referrals that reached a visit with 95% confidence interval; SUS mean; perceived confidentiality and help-seeking intention at baseline and end (paired t-test or Wilcoxon)
4. Stressors and reasons for leaving	Frequency of each stressor and each reason for thinking of leaving, by year of study and month; share reporting informal borrowing, security held and lost-card report (Prefer not to say counted separately); thematic analysis of stressors in own words; cross-tabulation with colour
5. Change and early exit	Paired t-test or Wilcoxon on PHQ-9, GAD-7, AUDIT-C and resilience, first versus latest check; association of the change with perceived consequences and honesty items; exploratory Firth logistic regression for early exit with the limits in 3.5

Objective	Analysis in jamovi
6. Cost and adoption	Cost per active student and per contact from the cost log; scale-up scenarios (Chapter Five)

Missing data will be described. Items skipped by design (adaptive items) are not treated as missing. Tests are two-sided at 5%; because this is a pilot, effect sizes with confidence intervals are emphasised over p-values. Small groups (for example one programme in one year) are never reported if fewer than ten people.

3.8 Open Suggestions and Future Adjustment

Because intensive methods such as group discussions are not used, the study keeps **room for opinion** in three ways: three open suggestion boxes in the check, a comment box in the end-of-pilot feedback, and a comment column in the expert review sheet. Every comment is read by the researcher and the steering group, coded into themes, and listed in a **suggestions register** with a decision (adopt now, adopt later, decline) and a reason. Adopted changes go in the change log with a new protocol version, so future adjustments are traceable.

3.9 Quality and Rigour

3.9.1 Expert review, pre-testing and validation of the questions

Standard scales keep their published wording and cut-offs. The PC-PTSD-5, AUDIT-C and Brief Resilience Scale have not been validated in Kenyan military or college groups, and the life-pressure, money, perceived-consequences, thoughts-of-leaving and readiness items are **locally developed and not yet validated**. They are therefore treated as descriptive, and cut-offs from them are not used to decide risk. (Life pressures can move a student from green to amber, which only widens the offer of support. The red band depends on the validated symptom and safety items alone.) Four steps are used:

1. **Expert review:** five DCHS staff (for example a nurse tutor, a counsellor or chaplain, a welfare officer, a lecturer and a clinician) rate each question for clarity and relevance. This replaces a Delphi panel at much lower cost.
2. **Cognitive interviews:** 10 to 15 students think aloud and say what each question means to them. A question is reworded if more than one in five misread it.
3. **Timing and length:** each form is timed and shortened if it exceeds the minutes shown to students.
4. **Reliability and skips:** Cronbach’s alpha of 0.70 or more is sought for scales with several items; a non-sensitive item skipped by more than 15% of students is reviewed. Locally developed scales are reported as exploratory.

3.9.2 Technical assurance

The platform has passed automated testing on a local test database: browser flows in demo mode and through to a test database, checks of role and privacy rules, and tens of thousands of comparisons of scores between the app and the database with no differences. It has **not** yet been run against a live hosted project, security-reviewed by an independent person, or used on real phones in the field. Before any real participant is enrolled, the following must be passed (go or no-go): review of access rules and the data flow by a person outside the project, such as a DCHS or NDU-K ICT officer; the data protection impact assessment; a test on at least five phone models (including low-cost Android and an iPhone) with and without signal; a drill of the offline queue and of database failure; and a test that the public key can insert but cannot read or delete. Any critical finding stops enrolment until it is fixed.

3.9.3 Reporting and reflexivity

Reporting follows STROBE for the observational parts and StaRI for the implementation study. The researcher is a DCHS student. A supervisor outside the nurse rota reviews the coding of open comments, and the researcher has no access to the identity of participants.

3.10 Governance of the Adaptive Service

A small **steering group** (the researcher, the supervisor, a clinical lead, a welfare officer, an ICT and data-protection officer, and a student representative) meets every two weeks during the pilot. It reviews the service log, group early warnings, the suggestions register and the change log. Any change to a cut-off, rule or message must be written, dated and given a reason, and changes that affect risk handling need clinical lead approval. Changes that affect what is collected go to the ethics committee as an amendment.

3.11 Cost Control

The design keeps cost low by using free hosting and survey tools, open-source software, existing nurses, the students' own phones, a poster and QR code, and printed copies only for those who need them. There are no travel costs for participants and no paid interviewers. The budget is in Chapter Five.

3.12 Fallback if Approvals or Hosting Are Slow

If approvals for the live call-back service take longer than planned, the study carries on with **the same questions as a survey** with the engagement features (micro-lessons, ask-the-nurse, peer champions) and help numbers, but no live call-back. If the web app cannot be hosted, the same questions run on **KoboToolbox** or on **printed copies**. The call-back pilot follows when approvals are complete. This keeps the study on time without lowering safety.

3.13 Limitations

- The pilot shows **uptake and feasibility, not clinical improvement**. It is too short and too small to show that mental health improved or that suicides were prevented.
- Self-report screening is open to under-reporting when students fear consequences. The study asks about perceived consequences and honesty, and the check is not seen by staff and is not used to judge students.
- Registry outcome linkage depends on a designated officer and on consent; without it, early exit cannot be studied and thoughts of leaving are used instead.
- Results from one college may not carry over to others, or to serving units.
- The call-back service is limited to cover hours; overnight nurse cover is not promised (3.4.4).
- The locally developed items are not validated, and the sample may fall below the ideal level (3.5).
- The platform has been tested on a local test database and has not been security-reviewed or used in the field (3.9.2).
- The researcher is a student at the college under study; steps to reduce this influence are in Chapter Four.

CHAPTER FOUR: ETHICAL CONSIDERATIONS

4.1 Approvals and Agreements

- **Ethics review** by an institutional research ethics committee recognised for the study (to be named at submission), before any student is approached.
- **KDF-ISERC approval**, required for research within the military establishment. For serving personnel researching in military establishments the approval carries no fee.
- **NACOSTI** research licence.
- **A letter of support from DCHS** (and the hospital, for the nurse and the referral route) stating that (a) answers and results are never used for marks, progression, discipline, sponsorship or posting; (b) lecturers, examiners and commanders do not see individual answers or who took part; (c) a nurse and a referral route are available; and (d) data are kept as set out in 4.8.
- **Data protection:** registration as required, a **data protection impact assessment**, and advice from the Office of the Data Protection Commissioner where needed (Data Protection Act, 2019).
- **Stretch components** (applicants, serving units, family members) need their own agreements and an ethics amendment before they start (3.3.2).

4.2 Ethical Principles

The study follows the Nuremberg Code, the Declaration of Helsinki and the Belmont Report, and the principles drawn from them: respect for persons, beneficence, non-maleficence, justice and confidentiality.

Principle	How it is met
Respect for persons (autonomy)	Protections shown before anything else; plain-language information sheet; tap-to-consent with version and time kept; every question can be skipped and sensitive questions offer Prefer not to say; withdrawal at any time; colour sharing only by choice and reversible
Beneficence	A private result, practical next steps, a nurse call-back with a stated time in cover hours, welfare help for money problems, and early warning that lets help come before a crisis
Non-maleficence	Safety protocol for risk; Quick Exit and neutral messages; no lecturer or commander sees individual data; no use for marks or discipline; trained nurses with supervision; no AI use for crisis text
Justice	Offered to every enrolled student in every programme and year; offline use and low data use; printed copies for those without phones; Kiswahili wording planned; non-discrimination under the Mental Health (Amendment) Act, 2022
Confidentiality	No names, admission numbers or phone numbers with answers; a private code; access by role; groups under ten hidden; the one limit explained up front

4.3 Protections Guaranteed Before Taking Part

Every student sees the same short list of protections **before** being asked to agree, on the first screen of the app, at the top of the printed information sheet, and at the start of the Kobo form. The list is: (1) you choose, and nothing will happen to you if you say no, skip a question or stop; (2) this is not an assessment and does not affect your marks, progression, discipline, sponsorship, posting or place; (3) no name, admission number, service number or phone number is stored with your answers; (4) staff cannot see you: lecturers, commanders and college managers never see individual answers, and results are reported for groups of 10 or more only; (5) your result is yours, and a colour goes to the support office only if you choose to share it, and you can withdraw it; (6) help is close, with help numbers on every result and a nurse who can contact you if you ask; (7) the study is approved and supervised, with the ethics approval number and the contacts shown; and (8) one limit: if you tell a nurse that you or someone else is in immediate danger, the nurse may need to act to keep people safe. Answers already sent cannot be traced back to you, so they cannot be removed one by one.

4.4 The Central Ethical Question: Voluntariness and Power among Enrolled Students

Students are in a dependent position: the college controls their marks, progression and, for some, sponsorship. This raises four concerns, each answered in the design.

1. **Coercion and valid consent.** A student who thinks lecturers or commanders will see the result is not free to say no. **Answer:** lecturers, examiners and commanders do not recruit and do not see who took part; the invitation says plainly that taking part makes no difference; the letter of support commits the college to this; and answers carry no name or admission number.
2. **Honest answers.** Students may under-report if they fear consequences, and selection-style faking is a known problem in high-stakes settings (Birkeland et al., 2006). **Answer:** the protections come first; the check is not an assessment; and the study asks students how much they fear consequences and how honest they were, so that the effect can be measured.
3. **Use of results.** Individual scores stay on the phone. Colour sharing is opt-in, shows a colour and a reference chosen by the student, can be withdrawn, and is read only by the support office, with every look-up recorded. **Answer:** the check is a **support tool, not a screening gate**. The study's recommendation to the college is that it is never used to select, grade or discipline students (see 4.5).
4. **Dual role.** The researcher is a student at the college. **Answer:** the researcher does not teach, mark or select, has no access to identities, and works under a supervisor and a data custodian outside the student's own class.

4.5 What if the College Wishes to Use Results for Admission or Discipline?

The study will advise against it, for three reasons that the policy brief will set out. It would conflict with Article 27 of the Constitution and with the Mental Health (Amendment) Act, 2022. Answers given when stakes are high are less honest, so a score would be a poor guide. And fear of consequences would lead fewer students to use the service. The study's recommendation is that the check is used **to offer support to students and not to choose among them**. The college decides for itself, and the study's findings on honesty and uptake will be given to it in full. Applicants are not part of the study; if the college later wishes to extend to applicants, that needs a separate ethics amendment.

4.6 Informed Consent

Each student receives an information sheet and consent statements in the Data Collection Tools Pack. In the app the student reads the same text and taps to agree; the app stores the consent version and time. Printed copies are consented by tick-box, without a signature, so that no name is recorded; a sealed box is used for returns. Optional choices (sharing a colour; registry outcome file) are separate from consent to take part. Participants are not paid. Students must be 18 or older.

4.7 Safety Protocol and Duty of Care

The **written safety protocol** (Appendix A, Part D) governs how risk is handled:

- **Help on screen at once:** any thought of self-harm or a red result shows help numbers immediately, with an offer of a same-day call-back.
- **Cover hours and time targets:** nurse cover is 07:00 to 21:00 with an after-hours route through the duty medical officer (3.4.4). Times below are counted within cover hours. Same-day contact within eight hours for red or urgent requests; a visit within 48 hours for high risk; routine contact within seven days; follow-up within 72 hours; three call attempts over two hours, then escalation to the duty medical officer. These are targets that the pilot will test and the steering group may revise.
- **The nurse** asks directly about suicide, plans and means; agrees a safety plan; asks about safety at home and threats from lenders; arranges a visit before the call ends.
- **Limits of confidentiality:** if a student is at immediate risk of serious harm to self or others, the nurse may need to involve others; this is told to the student at the start and in the information sheet.
- **Staff care:** same-day debrief, supervision and rest for nurses who take red-band calls.
- **Serious events** are reported to the ethics committee within 24 hours.

Reaching a student who is anonymous. Anonymity protects the research data; help works because the student chooses to be reachable when they ask. A student who asks for a call-back may type a safe contact detail; the nurse then sees a ticket code, the urgency and only

the scores the student chose to share, and replies by phone or in the app. A student may instead walk in and quote the ticket code. If a student shares no contact, the nurse cannot find them; help numbers show at once and the reply waits in the app. The consent screen says that a contact detail, once given, will be used by the nurse. A written **duty-of-care protocol** for red results (who is called, within how many hours, and what happens if there is no answer) is submitted with this proposal. The anonymous research survey on its own is untraceable, so it ends with help numbers and a separate link to the service that is not tied to the answers.

The service is a **support and screening tool, not a diagnosis**. Nothing on the screen replaces assessment by a clinician. Phone numbers shown in this proposal and in the app are **placeholders (+2547xxxxxxxx)**; they are replaced with the real nurse, duty officer and support-office numbers when the service is adopted, and national helplines are confirmed before go-live.

4.8 The Money, Loans and Held-Card Questions

The questions about informal borrowing, security held and lost-card reports are **sensitive** and **optional** with Prefer not to say. They ask about the situation and not about the lender: no lender's name, place or phone number is collected. They are used to guide welfare help (money counselling, SACCO, welfare officer, advice on a lost card). Answers do not change the colour and are never passed to college or command staff. Because holding an identity card as security and false loss reports may involve breaches of regulations, the study will agree with DCHS and the legal officer, **before** the pilot, on a **no-penalty route** for a student who asks for help with a card held by a lender, and the information sheet will say so. If no such route is agreed, those items are removed from the live service (the DISABLED_ITEMS setting does this without rebuilding the app).

4.9 Family and Violence Disclosures

Students may disclose violence at home or elsewhere. The app shows the gender violence helpline (1195), the emergency number and local contacts on the same screen as the check. The nurse asks about safety, offers a safety plan and referral to a service of the student's choice, and does not contact anyone without the student's agreement, except for imminent danger. Children at risk are handled under the law on child protection, which the information sheet states.

4.10 Data Protection and Hosting

- **Minimisation:** no names, admission numbers, service numbers or national ID numbers; phone numbers are collected only when a student asks for a call-back, are stored apart from the answers, and are deleted after follow-up closes (period set at ethics review).

- **Anonymous ID:** each student receives the next anonymous number in order of joining (for example STUD-0001), tied to no name; if there is no signal, the phone makes a random code.
- **Security:** encryption in transit; access by role; no shared logins; an access log for any identifiable look-up; weekly backups held in a secure place.
- **Hosting:** the survey and anonymous answers use free services (GitHub Pages and Supabase, or KoboToolbox) with the ethics committee's approval. The **live call-back service**, which holds contact numbers, will run in Kenya or keep a serving copy in Kenya (Data Protection (General) Regulations, 2021, regulation 26); DCHS or the Medical Services would then be requested to host it (3.4.5).
- **Retention:** de-identified research data kept for the period the committee sets, then deleted.
- **Small cells:** group figures are not shown for groups of fewer than ten.

4.11 Use of Artificial Intelligence

The default guide runs **on the phone** and answers from approved lessons. An **AI guide** is built as an option and is **off**. It will be used only if the ethics committee and data-protection review approve it, with these limits: it never diagnoses, never replaces a person, is told to refer on crisis and danger words, and does not receive text that the phone marks as crisis. If it is off or fails, the phone guide takes over. A nurse can be reached from every screen.

4.12 Benefits, Risks and Their Balance

Benefits to students include a private result, guidance and a quick route to a nurse; to the college, early sight of group-level strain and a costed plan; to the wider community, evidence for services. **Risks** are discomfort from sensitive questions, worry from a red result, loss of privacy if a phone is shared, and misuse of results. Each is reduced in 4.3 to 4.10. The remaining risk is judged low to moderate, and the benefit of reaching students early is judged to outweigh it.

4.13 Conflicts of Interest and Independence

The researcher is a student of DCHS and is supervised by DCHS staff. No payment is received from any technology supplier; the tools used are open-source or free to use. The college and the hospital do not review individual data and do not approve the findings. Results will be reported whether or not they favour the service.

CHAPTER FIVE: WORK PLAN, BUDGET AND THE CASE FOR INVESTMENT

5.1 Work Plan

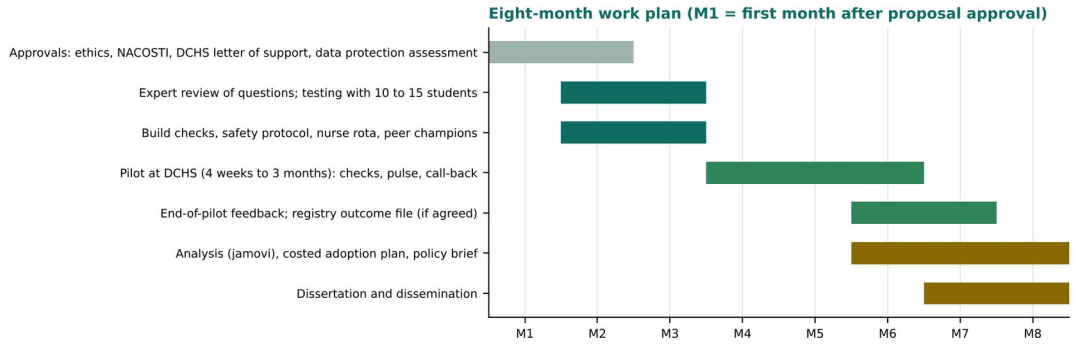


Figure 4. Eight-month work plan

5.2 Pilot Budget

The budget below is for the **research and pilot only**. Amounts are **estimates** made for this proposal and are to be confirmed against current prices and the rules of the bodies concerned. Nurse time and clinical supervision are expected as **in-kind** contributions of the hospital and DCHS unless otherwise agreed. There are no fees for participants and no paid interviewers.

Table 5.1. Estimated pilot budget (KES)

Item	Basis	Estimated cost (KES)
Printing: information sheets, printed questionnaires, posters, sealed boxes	About 100 printed copies and posters; the tools pack for the ethics committee	12,000
Airtime and mobile data: nurse call-backs, testing, reminders	Pilot period (up to 3 months)	15,000
Web hosting of the survey and database: GitHub Pages and Supabase free plans (or KoboToolbox free)	Free for the pilot. DCHS may be asked to host a Kenyan copy at adoption	0
Expert review and testing with students: refreshments and airtime	Five staff and 10 to 15 students	8,000
Peer champions: briefing and materials	Phase 1	5,000
Nurse on-call honorarium (3 months), if not provided in kind	Call-back service in cover hours	30,000
Second on-call nurse: training and airtime	Cover hours rota	5,000
Ethics review and NACOSTI fees (KDF-ISERC approval carries no fee)	To be confirmed with the bodies	25,000
Data protection impact assessment and independent security review	Phase 0 and 1; may be in kind from an ICT officer	15,000

Item	Basis	Estimated cost (KES)
Local transport in Nairobi	Briefings, meetings, supervisor	10,000
Printing and binding: dissertation, policy brief	Phase 3	15,000
Subtotal		140,000
Contingency (10%)		14,000
TOTAL		154,000

Hosting. The survey does not need the college to host it. The app is published free on GitHub Pages (or Netlify or Cloudflare Pages), the answers are kept in a free Supabase project, and KoboToolbox is a free fallback; the cost for the pilot is therefore **KES 0**. If the live call-back service later needs contact numbers, it must run in Kenya, and **DCHS or the Medical Services would then be requested to host a Kenyan copy** (the running cost is in Table 5.2). The software is open-source, so there is no licence cost.

5.3 Why Invest: The Case for Administration

The college and the Medical Services are asked to support a small, low-cost pilot because it offers four things they do not have today.

What the institution gains	How the pilot provides it
Early sight of problems, without naming anyone. Group-level early warning shows a rise in strain in a class or year group within a week, so leaders can act on workload, money advice or leave before people reach crisis.	Weekly pulse; group alerts for groups of ten or more; monthly report
A fast, confidential route to care. A person who asks for help hears from a nurse within a set time, and the time is measured by the database, not by memory.	Call-back clock; referral completion; safety protocol
A way to protect students and cut avoidable losses. Each student who leaves is a lost place and a lost investment in training. Early support in the first weeks is when help is most likely to change the outcome.	Student check; support-office colour (opt-in); outcome file
Evidence for a decision. A costed plan and a policy brief, based on use, time and cost in the college's own setting, rather than on foreign studies.	Service log; cost log; end-of-pilot feedback

5.3.1 The cost of doing nothing

No official figures were found on suicide, violence at home or debt among Kenya Defence Forces personnel and families. Reports on these issues in the media are **anecdotal** and are treated as signals, not as rates. The published evidence from other military groups shows that these problems are common: a pooled estimate of 12.7% for perpetration and 20.7% for

victimisation of intimate partner violence in military and veteran samples (Cowlishaw et al., 2022); about 26% past-year physical perpetration among serving and veteran personnel (Kwan et al., 2020); and about 31% with at least one mental health condition among 1,424 Kenyan university students (Mugotitsa et al., 2025), with family support linked to lower odds (adjusted odds ratio 0.47). Nursing programmes often lose about 10% to 30% or more of students (Sheikoleslami et al., 2025). The pilot will measure the local picture; until then, the college cannot see its own. A recommendation of the study, set out in the policy brief, is that the check is used to offer support and not to choose among applicants.

5.3.2 Costed adoption plan (illustrative)

If the pilot works, the next step is to run the service as part of routine support. The table below shows an **illustrative** running cost for DCHS for one year. **The assumptions are placeholders to be replaced with the college's own figures** (salary scales, number of users) and are not findings.

Table 5.2. Illustrative yearly running cost of the service at DCHS (KES)

Item	Estimated cost (KES)
Nurse call-back time (0.3 full-time equivalent at KES 90,000 a month)	324,000
Clinical supervision (5 hours a month at KES 3,000 an hour)	180,000
Hosting in Kenya and security upkeep	60,000
Airtime and data for call-backs and reminders	60,000
Peer champions and training refreshers	60,000
Part-time analyst for monthly group reports	100,000
TOTAL per year	784,000

Break-even. The service pays for itself if it helps the college keep enough students who would otherwise leave. The number needed equals the yearly running cost divided by the value of one retained place. The value of a place is **not known** and must be taken from the college's finance office; three illustrative values are shown.

Table 5.3. How many additional students must stay, each year, for the service to break even (illustrative)

Assumed value of one retained place (KES)	Students who must stay each year to break even
300,000	2.6
500,000	1.6
800,000	1.0

This shows the size of the question. It is **not a forecast**. If a place is worth KES 500,000, keeping about 2 students a year covers the running cost, before counting any benefit to

families, units or the lives protected by earlier care. The pilot will provide the real cost per active user and per contact, and the registry linkage will provide the real picture of who stays.

5.4 Early Intervention: Acting in Time to Avoid Extreme Outcomes

The service is designed so that signs are seen early and answered fast. Help is organised in tiers, each with a target time. The targets are set from the safety protocol and **will be tested in the pilot**.

Table 5.4. The escalation ladder and target times

Tier	Signal	What happens	Target time
0. Self-help	Green check; weekly pulse with no cue	Lessons, weekly pulse, buddy reminder	At once
1. Early cue	Pulse cue or amber (feeling the strain)	Tips, lessons, welfare navigator; offer of a nurse	Offer shown at once; call-back, if asked, within 24 hours
2. Struggling	Amber (struggling): PHQ-9 10–19, GAD-7 10 or more, positive trauma screen, alcohol, unsafe at home	Nurse call-back; confidential visit booked	Contact within 24 hours; visit within 7 days
3. High risk	Red; or any thought of self-harm	Help numbers at once; same-day call-back; clinician; safety plan	Contact within 8 hours (counted in cover hours); visit within 48 hours
4. Imminent danger	Danger described to the nurse or on call	Stay on the line; duty medical officer; emergency services; transport	Immediate
Group alert	Group of 10 or more rises 10 points above the average of the previous four periods	Aggregated alert to the welfare lead and medical officer; steering group reviews	Within 1 week

Lead time is the key measure: the number of hours from the first amber or red signal to first contact with a nurse, and from the request to a visit. The pilot will report the median and the share within target, and will compare it with the time that nurse and support-office records show it takes today for a student with a problem to be seen, so that the college can see how much earlier the service acts. The study does not claim that this lowers suicides or violence; it claims that it **shortens the time to help** and shows where the help does not reach.

5.5 Risk Register

Risk	Chance	Effect	Response
Approvals or agreement delayed	Medium	Pilot starts late	Fallback in 3.12; start with the survey version

Risk	Chance	Effect	Response
Low uptake (distrust of privacy)	Medium	Too few check-ins	Protections shown first; peer champions; class briefing; lecturers do not recruit; extend the pilot week by week
Nurse cannot meet call-back targets, or a request arrives out of hours	Medium	Unsafe delay	Stated cover hours; second nurse; after-hours route through the duty medical officer; pause enrolment if the rota cannot cope
Too few participants (below 100 check-ins)	Medium	Weak precision	Extend the pilot week by week; report counts and exact intervals only; frame as a pilot signal (Table 3.2)
Questions too long or misunderstood	Medium	Missing or poor data	Cognitive interviews; timing; shorten forms; reword items misread by more than one in five
Scope too large for eight months	Medium	Core unfinished	Core and stretch design (3.3.2); stretch parts never delay the core
Platform not yet security-reviewed or field-tested	Medium	Data or reliability problem	Go or no-go list (3.9.2) before any real participant
Result used for marks, progression or discipline	Low	Harm; loss of trust	Letter of support states this; staff cannot see answers; colour sharing is a student choice; access log
Privacy breach	Low	Serious harm	Access by role; encryption; breach plan; Kenyan copy for the live call-back service
Weak phone access, power or data	Medium	Lower reach	Offline use; low data use; Quick Exit; printed copies in a sealed box
Students under-report from fear of consequences	Medium	Biased scores	Protections shown first; anonymity; questions on perceived consequences and honesty; results never used to judge
Survey link cannot be hosted by the college or is blocked	Low	Delay	No college server needed (3.4.5); GitHub Pages or KoboToolbox; printed copies
Researcher bias (student of the college)	Medium	Influence on data	Independent supervisor; no access to identities; reporting of all results
Held-card items cause fear of penalty	Medium	Loss of trust	No-penalty route agreed in advance; items can be switched off

5.5.1 Dissemination and Expected Outputs

- A dissertation for the BSc Nursing with Military Health programme.
- A **two-page policy brief** and a briefing to the DCHS administration and the Medical Services.
- A costed adoption plan and the working service, with its guides, for use by other colleges of the National Defence University.
- A manuscript for a peer-reviewed journal, and an abstract for a nursing or military health conference.

- The de-identified dataset and codebook, if the ethics committee allows, for other researchers.

5.6 Sustainability and Scale-up

If the pilot meets its goals (use, trust and time targets), the plan is to (1) keep the service at DCHS as a routine support for enrolled students, and consider applicants as a later option; (2) extend to other colleges of the National Defence University; and (3) extend to units through the Medical Services, since students go to units after training and keep the same private code. Each step needs its own agreement, a trained nurse, and a hosting arrangement in Kenya (DCHS or the Medical Services would be requested to host). The tools are open-source and the question set is in a single file, so changes are cheap and are documented in the change log.

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APPENDICES

Appendix A: Data Collection Tools Pack

The full, printable pack (about 40 pages) is supplied as a separate file, **Data_Collection_Tools_Pack.docx**, for the ethics committee. It has the same words that students see in the app, and a printed copy can be used by students who prefer paper.

Part	Content
A	Your protections (shown before anything else); information sheet and consent for students
B1–B4	Questionnaires: DCHS student wellbeing check (with three open suggestion boxes); private check-in; weekly pulse; end-of-pilot feedback
C	Expert review sheet for the wording; how printed copies are handled; suggestions register
D	Safety protocol; colour bands and scoring; objectives-to-tools matrix; cost and service log template

Appendix B: Objectives-to-Tools Matrix

Objective	Tools	Analysis
1. Adapt the service and questions to DCHS students	Expert review sheet; testing with students; open suggestion boxes; end-of-pilot comments	Content validity index; timing; suggestions register
2. Run a pilot at DCHS	Service log; engagement counts; change log	Counts; run charts; fidelity
3. Measure uptake, time to contact, referral completion, usability, confidentiality, help-seeking	Usage events; call-back clock; baseline and end-of-pilot items	Medians, proportions, SUS, paired tests
4. Describe stressors named by students and reasons for thinking of leaving	Money items; stressor list; stressor in own words; weekly pulse cue; thoughts-of-leaving items	Frequencies by month and year group; thematic analysis
5. Describe change over the pilot; explore early exit	Student check at baseline and follow-up; consequences and honesty items; registry outcome file (if agreed)	Paired tests; exploratory Firth regression
6. Costed adoption plan and policy brief	Cost log	Cost per user and per contact; scale-up scenarios

Appendix C: Colour Bands

Colour	Meaning to the person	Rule (summary)	Offered
Green	Coping well	No flagged need (zone 1)	Lessons, weekly pulse, check again in about 4 weeks

Colour	Meaning to the person	Rule (summary)	Offered
Amber	Feeling the strain; struggling	Zones 2 and 3: mild or moderate symptoms, high life pressures, positive trauma or alcohol screen	Lessons, welfare navigator, nurse call-back
Red	Needs care now	Zone 4: any thought of self-harm, or severe symptoms	Help numbers at once; same-day call-back

Appendix D: Assumptions and Evidence Register

This register separates what is established, what is assumed and what still has to be checked, so that the panel can see how far each statement can be trusted.

Statement in the proposal	Status	Basis or action
Military personnel often do not seek help for mental health problems; stigma is a main barrier	Established	Sharp et al., 2015; Hoge et al., 2004
Intimate partner violence and mental health conditions are common in military and student groups	Established elsewhere; not measured in the KDF	Cowlishaw et al., 2022; Kwan et al., 2020; Mugotitsa et al., 2025
Suicide, family violence and debt are rising in the KDF community, and more students are leaving DCHS	No official figure found; media reports and staff concerns only	Treated as signals; the pilot measures the local picture, and the registry gives the leaving rate if it agrees
People answer more favourably when they fear consequences	Established for job-selection personality tests; assumed for mental health screens among students	Birkeland et al., 2006; consequences and honesty items measure it here
PHQ-9 and GAD-7 work in East Africa	Established	Gelaye et al., 2013; Mwangi et al., 2020
Life-pressure, money, thoughts-of-leaving, consequences and readiness items	Locally developed, not validated	Pre-test and reliability (3.9.1); descriptive use only
Expected share of users reaching a visit (20%)	Assumption	Table 3.2 gives the precision at other sample sizes
Cost of a lost college place; nurse salary; running costs (Tables 5.2 and 5.3)	Illustrative placeholders	To be replaced with DCHS and hospital figures
Time targets (8 hours, 48 hours, 7 days)	Proposed targets	Tested in the pilot; steering group may revise
The platform works as described	Tested on a local test database only	Go or no-go list before use (3.9.2)
Fit for deployment conditions after graduation	Assumption	To be tested with graduates posted to units in a later phase
The survey can be hosted free without the college	Standard practice for static sites and free survey tools; not yet done for this study	Steps in the setup guide; the college is asked for support, not hosting (3.4.5)

Appendix E: Platform and Data Package

Supplied with this proposal: the phone app and staff console (a folder of static files); the database scripts; a setup guide (about 20 minutes); a jamovi guide with worked analyses; KoboToolbox forms with the same questions; a share link and QR code for students; and practice data with invented values. The practice data are for rehearsal only and are not findings.